

INDEPENDENT QUANTITATIVE RESEARCH PROJECT

Systematic Cross-Sectional Equity Strategy

Machine-learning momentum ranking of S&P 500 constituents —
built, audited, and paper-trading live with a locked, append-only record.

234

out-of-sample months (2007-2026)

994

ticker universe incl. delisted

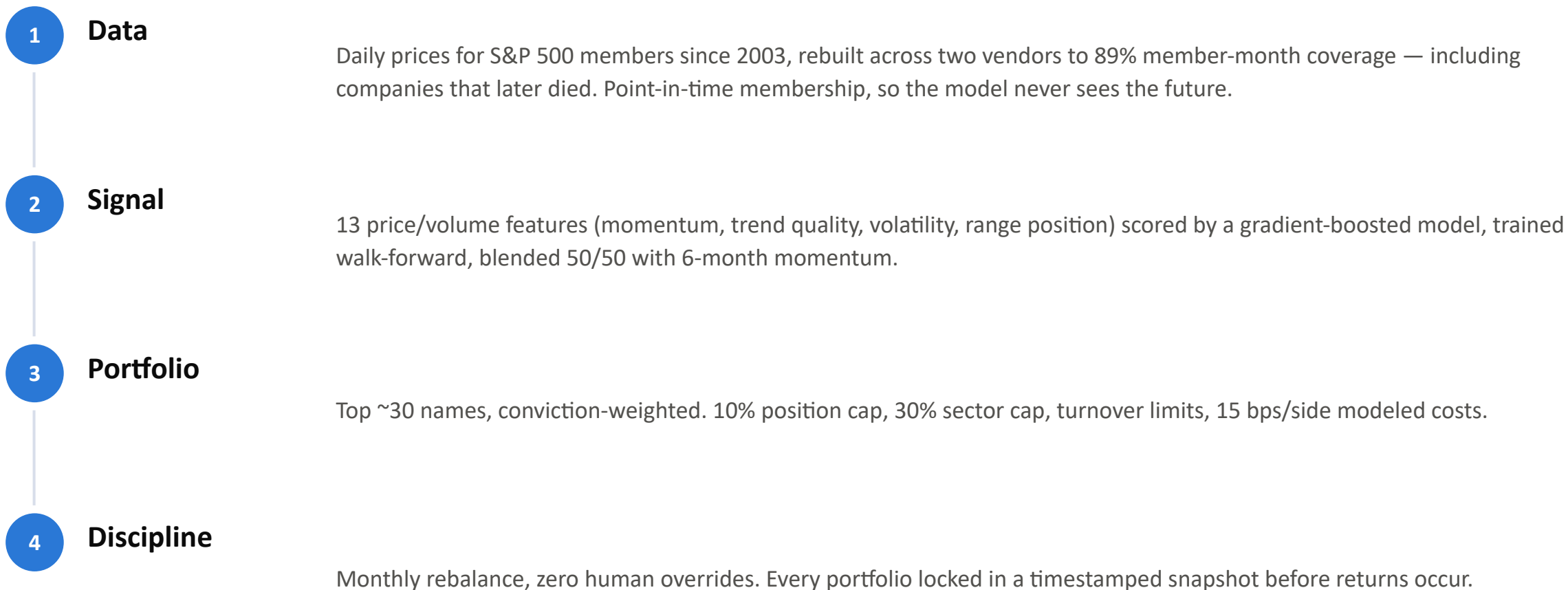
9 tested

hypotheses — 7 rejected

Zero-touch

automated live tracking

A disciplined momentum machine, rebuilt monthly



Data forensics before any modeling

MEMBERS LACKING DATA

31%

240 of 777 test-window members — $\approx$ a quarter of member-months, mostly losers

COVERAGE AFTER REBUILD

74% $\rightarrow$ 89%

aliases + two-vendor recovery, 365 identifiers

CORRUPT SERIES CAUGHT

12

mechanical gates, every exclusion logged



Survivorship bias — Free data deletes dead companies. Rebuilt the universe from a 994-ticker membership union with rename mapping (ANTM $\rightarrow$ ELV, PCLN $\rightarrow$ BKNG) and delisted-price recovery.



Ticker-reuse contamination — Found price series continuing past corporate death (e.g. “BBBY” quoting in 2026). Membership-window truncation removes them mechanically, no judgment calls.



Honest accounting — Dividends included, delisting exits charged, 15 bps/side costs, execution-timing sensitivity measured at approximately zero.

Every honesty fix lowered the backtest — 25.7% $\rightarrow$ 20.1% $\rightarrow$ 15.0% $\rightarrow$ 11.0% annualized. The final fix erased the edge; the truth got complete.

Results — and what they honestly mean

ANNUALIZED

11.0%

S&P 500: 10.8%

CAPM ALPHA

-0.4%/yr

$t = -0.15 \cdot \text{beta } 1.14 \cdot \text{no edge}$

SHARPE

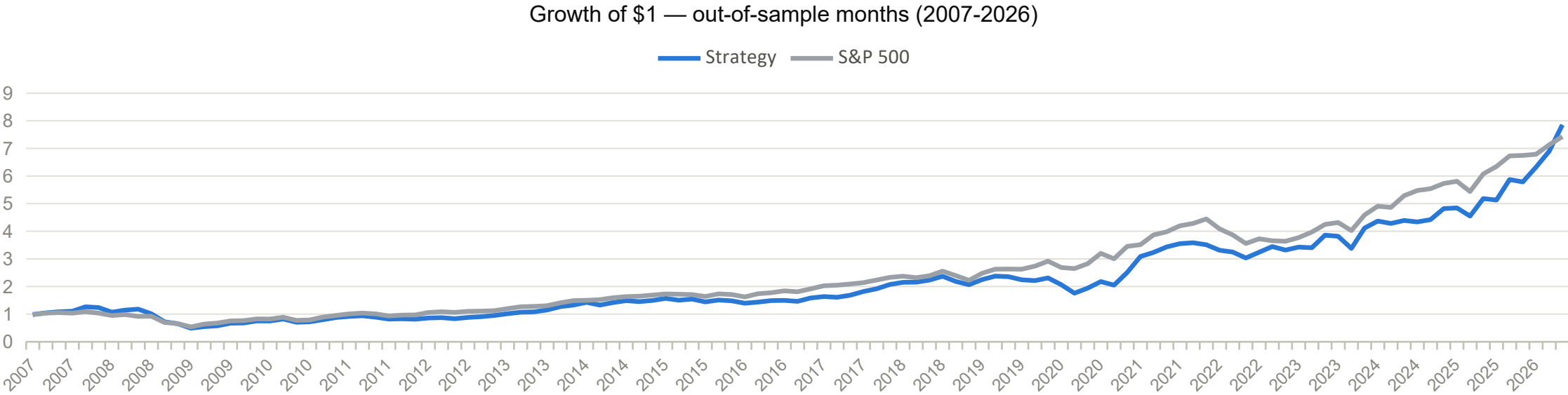
0.51

benchmark 0.70 · return/vol, same convention

HIT RATE

51.7%

of months vs SPY — a coin flip



Ten of twenty years lag the index — a coin flip



On complete data the strategy lags the index in 10 of 20 years. That symmetry — with hit rate 51.7% of months — is exactly what alpha of approximately zero looks like. On the earlier, incomplete data the same chart appeared to show a persistent edge.

Nine hypotheses, pre-registered rules, seven rejections

Conviction (score) weighting	ADOPTED	+1.4pp/yr, Sharpe up, beta flat — confirmed on three samples
10% position cap	ADOPTED	Zero return cost; max single-name risk 12.9% → 10%
Concentration (top 10-20)	REJECTED	Beta in disguise (1.16 → 1.31 in that test's pre-adoption config); Sharpe fell
Inverse-volatility weights	REJECTED	Cut the excess return roughly in half
Volatility-target overlay	REJECTED	Negative alpha on the tested sample
Fundamental factors ×3 (SEC EDGAR)	REJECTED	Profitability $t \approx 0.55$; accruals sign flipped; asset growth wrong sign
Extra momentum features	REJECTED	Alpha 3.1% → 0.0% — correlated features fed the model noise
Execution timing (robustness check)	VERIFIED	Next-open vs close: return unchanged — implementable as modeled
Complete-data retest of adopted choices	SUPERSEDED	At 89% coverage no variant retains alpha — the gains were survivorship artifacts

Acceptance rules were committed before results were seen — and the final retest was allowed to overturn even the accepted answers.

Four things the data taught me

Survivorship is the master bias

An apparent +4pp/yr edge — alpha 3.1, t 1.19 — was eliminated entirely by recovering the dead companies free data had deleted. Stripping bias comes before stripping beta.

Edges are coin leans — and fragile

The portfolio's 55% month-win rate on incomplete data fell to 51.7% on complete data: the coin's lean was borrowed from the missing losers.

More features can subtract

Adding two plausible momentum features destroyed the alpha entirely. With weak signals, model capacity is a liability, not an asset.

Classic factors have decayed

Profitability, accruals and asset growth — decades of citations — no longer clear a pre-registered gate in US large caps. Published edges get arbitrated.

A track record no backtest can fake

1 Locked snapshots

Each month's portfolio is frozen in a timestamped, append-only file before its returns occur — OneDrive version history provides the tamper evidence.

2 Paper brokerage

A simulated \$100k account executes whole-share orders at month-end closes with fees; full blotter, positions, and P&L.

3 Scored ledger

Every completed month graded at real prices vs SPY, net of costs — one row per month, never edited.

4 Zero-touch automation

Scheduled runs (monthly rebalance + daily marks) with failure tripwires and a written operations manual. No human in the loop.

Judgment at 12 ledger rows; decisions at 24. Until then the protocol is: run, record, do not touch.

What this is not — and where it goes next

KNOWN LIMITATIONS

- Alpha $-0.4\%/yr$ ($t = -0.15$) on complete data: no demonstrable edge — index-like returns at $1.14\times$ the risk
- Beta 1.14 — falls harder than the index in crashes (2008: -37% vs -27%)
- One signal family (momentum); fundamentals failed the gate
- 11% of member-months remain unobtainable; the $79\% \rightarrow 89\%$ completion moved returns $-4pp/yr$, so the residual gap is a real caveat on an already-null result

NEXT

- Accumulate the live record — it now tests the null result out of sample
- Mid-cap universe: a preliminary cross-market test (survivorship-biased data, haircut estimate $\sim 3-4pp/yr$ vs $\sim 2pp$ here) suggests a larger edge — to be validated under the same gates
- More markets, not more parameters — breadth is the honest way to scale a weak signal

THE POINT

The result isn't the return. It's the process that survives scrutiny.

Point-in-time data,

rebuilt and audited

Pre-registered tests,

most of them failed

Honest statistics,

alpha claimed only with its t-stat

Live evidence,

accumulating untouched